

BIEK COMPUTER SCIENCE LECTURES

BIEK Computer Science XII

One PDF per lecture — plus a full chapter pack for each unit.

Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — بارہویں جماعت

XII

Lecture index

Chapter packs are named **Lecture 01–06**. Individual topics are **ChNN-LNN**. ★ marks Golden (high-yield) exam topics.

Code	File	Title
Lecture 01	CS-XII-Lecture-01-Computer-Systems-HCI.pdf	Computer Systems (HCI)
Ch01-L01	CS-XII-Ch01-L01-1.1-Fundamentals-of-Human-Computer-Interaction-HCI.pdf	1.1 Fundamentals of Human Computer Interaction (HCI)
Ch01-L02	CS-XII-Ch01-L02-1.1.1-1.1.2-Traditional-vs-Natural-Interaction.pdf	1.1.1 – 1.1.2 Traditional vs Natural Interaction
Ch01-L03	CS-XII-Ch01-L03-1.2-Applications-of-HCI-Domains-of-Life.pdf	1.2 Applications of HCI (Domains of Life) ★
Ch01-L04	CS-XII-Ch01-L04-1.3-Components-of-HCI.pdf	1.3 Components of HCI
Ch01-L05	CS-XII-Ch01-L05-1.3-Types-of-User-Interaction.pdf	1.3 Types of User Interaction
Ch01-L06	CS-XII-Ch01-L06-1.3-Interface-Types-Environment-Feedback.pdf	1.3 Interface Types, Environment & Feedback
Ch01-L07	CS-XII-Ch01-L07-1.4-Importance-of-HCI.pdf	1.4 Importance of HCI ★
Ch01-L08	CS-XII-Ch01-L08-1.5-Accessibility-Principles.pdf	1.5 Accessibility Principles
Ch01-L09	CS-XII-Ch01-L09-1.6-Need-Analysis-for-Interface-Design.pdf	1.6 Need Analysis for Interface Design
Ch01-L10	CS-XII-Ch01-L10-1.7-Human-Computer-Interaction-Problems.pdf	1.7 Human Computer Interaction Problems ★
Ch01-L11	CS-XII-Ch01-L11-1.8-Methods-for-Improving-HCI.pdf	1.8 Methods for Improving HCI
Ch01-L12	CS-XII-Ch01-L12-1.9-User-Interface-Design-UI-vs-UX.pdf	1.9 User Interface Design (UI vs UX) ★
Ch01-L13	CS-XII-Ch01-L13-1.9.3-1.9.5-Wireframing-Figma-and-Prototypes.pdf	1.9.3 – 1.9.5 Wireframing, Figma, and Prototypes
Ch01-L14	CS-XII-Ch01-L14-1.11-Test-and-Evaluate-HCI-Evaluation-Methods.pdf	1.11 Test and Evaluate HCI (Evaluation Methods)

Ch01-L15	CS-XII-Ch01-L15-1.11.2-Testing-Methods-Usability-AB-Automated.pdf	1.11.2 Testing Methods (Usability, A/B, Automated) ★
Lecture 02	CS-XII-Lecture-02-Computational-Thinking-Algorithms.pdf	Computational Thinking & Algorithms
Ch02-L01	CS-XII-Ch02-L01-2.1-Analyze-algorithms-for-correctness.pdf	2.1 Analyze algorithms for correctness
Ch02-L02	CS-XII-Ch02-L02-2.1.1-2.1.3-Trace-Tables-vs-Stepwise-Reasoning.pdf	2.1.1 – 2.1.3 Trace Tables vs Stepwise Reasoning ★
Ch02-L03	CS-XII-Ch02-L03-2.2-Evaluating-the-Clarity-of-an-Algorithm.pdf	2.2 Evaluating the Clarity of an Algorithm
Ch02-L04	CS-XII-Ch02-L04-2.2.1-2.2.2-Modularity-and-Readability.pdf	2.2.1 – 2.2.2 Modularity and Readability
Ch02-L05	CS-XII-Ch02-L05-2.3-Assess-Algorithm-Efficiency.pdf	2.3 Assess Algorithm Efficiency ★
Ch02-L06	CS-XII-Ch02-L06-2.3.1-Number-of-Steps-Big-O-Notation.pdf	2.3.1 Number of Steps (Big O Notation) ★
Ch02-L07	CS-XII-Ch02-L07-2.3.2-Number-of-Conditions.pdf	2.3.2 Number of Conditions
Ch02-L08	CS-XII-Ch02-L08-2.3.3-Number-of-Repetitions.pdf	2.3.3 Number of Repetitions
Ch02-L09	CS-XII-Ch02-L09-2.3.4-The-Efficiency-Evaluation-Framework.pdf	2.3.4 The Efficiency Evaluation Framework
Ch02-L10	CS-XII-Ch02-L10-2.4-Refinements-to-improve-clarity-and-efficiency.pdf	2.4 Refinements to improve clarity and efficiency
Ch02-L11	CS-XII-Ch02-L11-2.5-Concept-of-Data-Structure.pdf	2.5 Concept of Data Structure ★
Ch02-L12	CS-XII-Ch02-L12-2.5.1-Linear-Data-Structures-Array-Linked-List-Stack-Queue.pdf	2.5.1 Linear Data Structures (Array, Linked List, Stack, Queue) ★
Ch02-L13	CS-XII-Ch02-L13-2.5.2-Non-Linear-Data-Structures-Tree-Graph.pdf	2.5.2 Non-Linear Data Structures (Tree, Graph)
Ch02-L14	CS-XII-Ch02-L14-2.5.3-Operations-on-Data-Structures.pdf	2.5.3 Operations on Data Structures

Ch02-L15	CS-XII-Ch02-L15-2.6-Identify-data-structures-in-problem-scenarios.pdf	2.6 Identify data structures in problem scenarios
Ch02-L16	CS-XII-Ch02-L16-2.7-Trace-data-retrieval-in-Array-List-and-Queue.pdf	2.7 Trace data retrieval in Array, List and Queue
Lecture 03	CS-XII-Lecture-03-Programming-Fundamentals.pdf	Programming Fundamentals
Ch03-L01	CS-XII-Ch03-L01-3.1-Data-Structures-in-Python.pdf	3.1 Data Structures in Python
Ch03-L02	CS-XII-Ch03-L02-3.1.1-Lists-Ordered-Mutable-Methods.pdf	3.1.1 Lists (Ordered, Mutable, Methods) ★
Ch03-L03	CS-XII-Ch03-L03-3.1.2-Tuples-Ordered-Immutable.pdf	3.1.2 Tuples (Ordered, Immutable) ★
Ch03-L04	CS-XII-Ch03-L04-3.1.3-Sets-Unordered-Unique-Math-Operations.pdf	3.1.3 Sets (Unordered, Unique, Math Operations)
Ch03-L05	CS-XII-Ch03-L05-3.1.4-Dictionaries-Key-Value-pairs.pdf	3.1.4 Dictionaries (Key-Value pairs) ★
Ch03-L06	CS-XII-Ch03-L06-3.1.5-Common-Built-in-Functions-min-max-len-sum.pdf	3.1.5 Common Built-in Functions (min, max, len, sum)
Ch03-L07	CS-XII-Ch03-L07-3.1.6-Activity-Students-Attendance-Program.pdf	3.1.6 Activity: Students Attendance Program
Ch03-L08	CS-XII-Ch03-L08-3.1.7-Activity-Word-Frequency-Counter.pdf	3.1.7 Activity: Word Frequency Counter
Ch03-L09	CS-XII-Ch03-L09-3.2-Functions-in-Python.pdf	3.2 Functions in Python ★
Ch03-L10	CS-XII-Ch03-L10-3.2.1-Types-of-Functions-Built-in-vs-User-defined.pdf	3.2.1 Types of Functions (Built-in vs User-defined)
Ch03-L11	CS-XII-Ch03-L11-3.2.2-Functions-with-Return-Values.pdf	3.2.2 Functions with Return Values
Ch03-L12	CS-XII-Ch03-L12-3.2.3-Activity-Arithmetic-Calculator-Using-Functions.pdf	3.2.3 Activity: Arithmetic Calculator Using Functions
Ch03-L13	CS-XII-Ch03-L13-3.2.4-Scope-of-Local-and-Global-Variables.pdf	3.2.4 Scope of Local and Global Variables

Ch03-L14	CS-XII-Ch03-L14-3.3-3.4-File-Handling-and-Operations-Modes-r-w-a.pdf	3.3 & 3.4 File Handling and Operations (Modes r, w, a) ★
Ch03-L15	CS-XII-Ch03-L15-3.6-with-Statement-Context-Manager.pdf	3.6 with Statement (Context Manager) ★
Ch03-L16	CS-XII-Ch03-L16-3.8-Errors-and-Exceptions-Try-Except.pdf	3.8 Errors and Exceptions (Try-Except)
Ch03-L17	CS-XII-Ch03-L17-3.9-Mini-File-Handling-Project-Grade-Tracker.pdf	3.9 Mini File Handling Project: Grade Tracker
Lecture 04	CS-XII-Lecture-04-Data-and-Analysis.pdf	Data and Analysis
Ch04-L01	CS-XII-Ch04-L01-4.1-Concept-of-Data-Analysis.pdf	4.1 Concept of Data Analysis
Ch04-L02	CS-XII-Ch04-L02-4.1.1-4.1.2-Data-Sources-and-Database-Connection.pdf	4.1.1 – 4.1.2 Data Sources and Database Connection ★
Ch04-L03	CS-XII-Ch04-L03-4.2-Creating-a-SQLite-Database-in-Python.pdf	4.2 Creating a SQLite Database in Python
Ch04-L04	CS-XII-Ch04-L04-4.3-Basic-Operations-on-Data-using-Pandas.pdf	4.3 Basic Operations on Data using Pandas ★
Ch04-L05	CS-XII-Ch04-L05-4.3.3-Handling-Missing-Values-NaN.pdf	4.3.3 Handling Missing Values (NaN) ★
Ch04-L06	CS-XII-Ch04-L06-4.4-Data-Organization-and-Representation.pdf	4.4 Data Organization and Representation
Ch04-L07	CS-XII-Ch04-L07-4.5-Data-Visualization-Types-of-Graphs.pdf	4.5 Data Visualization & Types of Graphs ★
Ch04-L08	CS-XII-Ch04-L08-4.6-Descriptive-Statistics.pdf	4.6 Descriptive Statistics ★
Lecture 05	CS-XII-Lecture-05-Application-and-Impacts-of-Computing.pdf	Application and Impacts of Computing
Ch05-L01	CS-XII-Ch05-L01-5.1-Machine-Learning-Neural-Networks-Deep-Learning.pdf	5.1 Machine Learning, Neural Networks & Deep Learning
Ch05-L02	CS-XII-Ch05-L02-5.1.1-Machine-Learning-ML.pdf	5.1.1 Machine Learning (ML)
Ch05-L03	CS-XII-Ch05-L03-5.1.2-Neural-Networks-NN.pdf	5.1.2 Neural Networks (NN) ★

Ch05-L04	CS-XII-Ch05-L04-5.1.3-Deep-Learning-DL.pdf	5.1.3 Deep Learning (DL)
Ch05-L05	CS-XII-Ch05-L05-5.1.4-Components-of-Neural-Networks.pdf	5.1.4 Components of Neural Networks ★
Ch05-L06	CS-XII-Ch05-L06-5.1.5-Components-of-Deep-Learning.pdf	5.1.5 Components of Deep Learning
Ch05-L07	CS-XII-Ch05-L07-5.2-Applications-of-Neural-Networks-and-Deep-Learning.pdf	5.2 Applications of Neural Networks and Deep Learning
Ch05-L08	CS-XII-Ch05-L08-5.3-Secure-Collaboration-Authentication-Access-Control.pdf	5.3 Secure Collaboration (Authentication, Access Control) ★
Ch05-L09	CS-XII-Ch05-L09-5.3.2-Data-Protection-Encryption-Passwords-Backup-Firewall.pdf	5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)
Ch05-L10	CS-XII-Ch05-L10-5.4-Security-Threats-and-Mitigation-Techniques.pdf	5.4 Security Threats and Mitigation Techniques ★
Ch05-L11	CS-XII-Ch05-L11-5.4.2-Identify-Security-Threats.pdf	5.4.2 Identify Security Threats
Ch05-L12	CS-XII-Ch05-L12-5.4.3-Apply-Mitigation-Techniques.pdf	5.4.3 Apply Mitigation Techniques
Ch05-L13	CS-XII-Ch05-L13-5.5-Equity-and-Equal-Access-in-Digital-Collaboration.pdf	5.5 Equity and Equal Access in Digital Collaboration
Ch05-L14	CS-XII-Ch05-L14-5.5.1-Functions-of-Equity-and-Equal-Access.pdf	5.5.1 Functions of Equity and Equal Access ★
Ch05-L15	CS-XII-Ch05-L15-5.5.2-Collaboration-Tools.pdf	5.5.2 Collaboration Tools
Lecture 06	CS-XII-Lecture-06-Entrepreneurship-in-the-Digital-Age.pdf	Entrepreneurship in the Digital Age
Ch06-L01	CS-XII-Ch06-L01-6.1-Entrepreneur-and-Entrepreneurship.pdf	6.1 Entrepreneur and Entrepreneurship
Ch06-L02	CS-XII-Ch06-L02-6.2-Entrepreneurship-in-the-Digital-Age.pdf	6.2 Entrepreneurship in the Digital Age ★

Ch06-L03	CS-XII-Ch06-L03-Table-6.1-Local-examples-of-digital-entrepreneurship.pdf	Table 6.1 Local examples of digital entrepreneurship
Ch06-L04	CS-XII-Ch06-L04-6.3-From-Problem-to-Business-Idea.pdf	6.3 From Problem to Business Idea
Ch06-L05	CS-XII-Ch06-L05-6.4-Prototype.pdf	6.4 Prototype ★
Ch06-L06	CS-XII-Ch06-L06-6.5-Types-of-Prototypes-Low-Mid-High-Fidelity.pdf	6.5 Types of Prototypes (Low, Mid, High-Fidelity)
Ch06-L07	CS-XII-Ch06-L07-6.6-Prototype-Development-Cycle-Design-Build-Test-Iterate.pdf	6.6 Prototype Development Cycle (Design, Build, Test, Iterate) ★
Ch06-L08	CS-XII-Ch06-L08-6.7-Class-Activity-Create-a-Prototype.pdf	6.7 Class Activity: Create a Prototype
Ch06-L09	CS-XII-Ch06-L09-6.8-Minimum-Viable-Product-MVP.pdf	6.8 Minimum Viable Product (MVP) ★
Ch06-L10	CS-XII-Ch06-L10-6.9-Difference-between-Prototype-and-MVP.pdf	6.9 Difference between Prototype and MVP
Ch06-L11	CS-XII-Ch06-L11-6.10-Identifying-the-Riskiest-Assumption.pdf	6.10 Identifying the Riskiest Assumption
Ch06-L12	CS-XII-Ch06-L12-6.11-Developing-an-MVP.pdf	6.11 Developing an MVP
Ch06-L13	CS-XII-Ch06-L13-6.12-Case-Study-MVP-for-a-Canteen-Pre-Order-System.pdf	6.12 Case Study: MVP for a Canteen Pre-Order System
Ch06-L14	CS-XII-Ch06-L14-6.13-Testing-an-MVP.pdf	6.13 Testing an MVP
Ch06-L15	CS-XII-Ch06-L15-6.14-Beachhead-Market.pdf	6.14 Beachhead Market ★
Ch06-L16	CS-XII-Ch06-L16-6.15-6.16-Complete-Student-Project-Ethical-Use.pdf	6.15 & 6.16 Complete Student Project & Ethical Use
Lecture 07	CS-XII-Lecture-07-Final-Revision-and-Mock-Paper.pdf	Final Revision and Mock Paper